

HU CHONG XERN

hu.chongxern@u.nus.edu | 8861 1799 | <https://linkedin.com/in/chongxern-hu/> | <https://github.com/ChongXern>

Professional Summary

Final-year Computer Engineering undergraduate at NUS specialising in data and systems engineering. Builds end-to-end pipelines and software systems that transform ambiguous real-world problems into structured, usable solutions across IoT, machine learning and data workflows.

Education

National University of Singapore

August 2022 – Present

Bachelor of Engineering in Computer Engineering (Second Class Upper Honours)
Specialisation in Internet of Things

Boston University, United States

August 2024 – December 2024

Overseas Student Exchange Programme

Skills

Proficient Languages: Python, C/C++, Java, JavaScript, SQL

Frameworks/Tools: FastAPI, Springboot, Supabase, OpenCV, PyTorch, Power BI, Docker

Domains: Data Engineering, Software Systems, Machine Learning, Internet of Things

Work Experience

Data Analyst with Machine Learning at Henkel Singapore

January 2025 – June 2025

- Built and deployed a Python data pipeline packaged as an executable, enabling non-technical users across 7 APAC manufacturing sites to run asset synchronisation analysis independently.
- Designed algorithms to detect synchronised asset behaviour and identify opportunity timestamps from historical energy data between 2021 to 2024, uncovering operational patterns and anomalies.
- Built Power BI dashboards and presented insights to stakeholders, contributing to site-level operational reviews including the Bangpakong plant.

Software Quality Assurance Internship at Integro Technologies

May 2024 – July 2024

- Investigated production issues by tracing SQL logic, test cases and system behaviour for banking clients using Oracle systems.
- Reproduced bugs and collaborated with developers to identify root causes during QA testing.

Project Experience

IoT Greenhouse Final Year Project

September 2025 – April 2026

- Led end-to-end development of an IoT greenhouse system integrating sensors, computer vision, and backend pipelines for remote monitoring
- Developed a hybrid CV/ML pipeline using OpenCV and PyTorch to assess plant health from images, with a total of 130 labelled and pre-processed samples, gaining strong real-world performance metrics.
- Engineered data flow from image capture to storage and inference using Supabase, supporting continuous monitoring across 6 sensor nodes.
- Built a full-stack IoT monitoring and control dashboard using FastAPI and React, enabling real-time sensor streaming, greenhouse live interactive visualisation and remote device control, backed by scalable Supabase storage.

Embedded Computer Vision Motion Tracking Robotic System

September 2025 – November 2025

- Deployed an AprilTag-based computer vision system using ESP32-CAM to facilitate real-time user tracking and distance computation for an autonomous mobile robot.
- Created a multi-task Convolutional Neural Network (CNN) for combined binary classification and 8-value corner regression, improving model accuracy by applying fine-tuning and custom data augmentation.
- Managed the end-to-end process of embedded camera integration, including low-latency frame streaming from camera module to the main server.

NLP Project Allocation Clustering

April 2025 – May 2025

- Developed unsupervised NLP pipeline using text embeddings and clustering to categorise project proposals for Henkel, simplifying task allocation across hundreds of entries.
- Processed and analysed hundreds of textual project descriptions for task allocation and prioritisation.

FinTrack Financial Management Application

March 2024 – April 2024

- Developed Java-based financial management application supporting income, expense, and transaction tracking.
- Designed modular command-driven architecture using object-oriented principles, implementing structured transaction management with separate inflow/outflow abstractions.
- Built features for transaction editing, searching, categorisation and reporting, including an undo functionality for reversible operations.